Artificial Intelligence Engineering
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تحكم آلي
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LECTURE - (2)
موديل

قابلة للتغيير

لنظم التحكم

DYNAMICAL MODELS OF PHYSICAL SYSTEMS

10/23/2025

Note:- In Automatic controle system we must do simulation model first and then to reality implementing.

Outline

- Introduction
- Ordinary differential equations (ODEs)
- Time varying and time invariant systems
- Dynamical models
  - Electrical systems
  - Mechanical systems

المعادلات التفاضلية الاعتيادية / الاعتيادية
الأنظمة المتعلقة بالزمن والغير متغير بالزمن.

الانظمة الكهربائية
الانظمة الميكانيكية

تحليل تصميم أنظمة التحكم

- One of the most important tasks in the analysis and design of control systems is mathematical **modeling of system** subcomponents and ultimately the overall system.

المواد الثانوية ، بصورة فائقة

- The models of these systems are represented by **differential equations**, which may be linear or nonlinear.

خطية غير خطية تمثل معادلات تفاضلية

- We consider systems modeled by **ordinary differential equations** as opposed to partial differential equations.

المعادلات التفاضلية الاعتيادية وليس مركبة

ميكانيكي

- A control system may be composed of several components including mechanical, thermal, fluid, pneumatic, and electrical systems.

حراري سوائل ضغط الهواء الطاقة الكهربائية

- We review basic properties of some of systems, otherwise **known as dynamic systems.**

Ordinary differential equations (ODEs)

- First order differential equation (=the simplest dynamical model):

معادلة من الدرجة الأولى

$$\begin{cases} \dot{x}(t) = ax(t) & a \in \mathbb{R}, \quad \dot{x} \triangleq \frac{dx}{dt} \\ x(0) = x_0 & x_0 \in \mathbb{R} \end{cases}$$

- Its unique solution is: $x(t) = e^{at}x_0$

من الرتبة n بدون مدخل

- Differential equations of order n without input:

رتبة المشتقة معامل ثابت → output

$$\frac{dy^{(n)}(t)}{dt^n} + a_{n-1}\frac{dy^{(n-1)}(t)}{dt^{n-1}} + \cdots + a_1\dot{y}(t) + a_0y(t) = 0$$

- Differential equations of order n with input:

يتم تحويلها الصيغة الجبرية بلابلاس Laplace

$$\frac{dy^{(n)}(t)}{dt^n} + a_{n-1}\frac{dy^{(n-1)}(t)}{dt^{n-1}} + \cdots + a_1\dot{y}(t) + a_0y(t)$$
$$= b_{n-1}\frac{du^{(n-1)}(t)}{dt} + b_{n-2}\frac{du^{(n-2)}(t)}{dt} + \cdots + b_1\dot{u}(t) + b_0u(t)$$

→ Time Varying

→ Time Invarying

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- **Time varying and time invariant systems**

- This classification is based on whether the parameters of a system remain functions of time or are independent of time variables during operation.

- It is known that continuous dynamical systems are characterized by differential equation(s).

  - Time invariant system: A dynamical system is classified as time invariant if the coefficients of the differential equation(s) characterizing the system **are constants.**

ثوابت

  - Time varying systems: if any one, more than one, or all coefficients of the differential equation(s) characterizing the system are functions of time.

Dynamical models

أي عنصر تم دراسة تغير الزمن (متحول).

- A dynamical system is an object (or a set of objects) that evolves over time, possibly under external excitations.
- Examples: a car, a robotic arm, a population of animals, an electrical circuit, a portfolio of investments, etc.
- The way the system evolves is called the dynamics of the system.
- A dynamical model of a system is a set of mathematical laws explaining in a compact form and in quantitative way how the system evolves over time, usually under the effect of external excitations.

التطور الذي

تأثيرات خارجية

- Main questions about a dynamical system:
  - Understanding the system ("How X and Y influence each other ?") فهم النظام
  - Simulation ("What happens if I apply action Z on the system ?") المحاكاة
  - Design ("How to make the system behave the way I want ?") التصميم

Dynamical models

التجارب محدوديات

- Experiments provide an answer, but have limitations:
  - غالية جداً Maybe too expensive (example: launch a space shuttle)
  - خطرة جداً Maybe too dangerous (example: a nuclear plant)
  - ممكن ان تكون مستحيلة Maybe impossible (the system doesn't exist yet!)
- In contrast, mathematical models allows us to:
  - Capture the main phenomena (الظاهرة) that take place in the system (example: Newton's law – a force on a mass (كتلة) produces an acceleration (تعجيل / تسارع)).
  - Analyze the system (relations among dynamical variables).
  - Simulate the system (=make predictions (توقعات)) about how the system behaves under certain conditions and excitations (in analytical form, or on a computer).

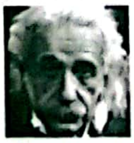
Dynamical models

لا يكلف !!

- Working on a model has almost zero cost compared to real experiments (just mathematical thinking, paper writing, computer coding).
- However, a simulation (or any other inference obtained from the model) is as better as the dynamical model is closer to the real system.
- Conflicting objectives:
  - موصوف او توصيف يشكل كامل Descriptive enough to capture the main behavior of the system
  - Simple enough for analyzing the system

  بسيط تحليل النظام

> "Make everything as simple as possible, but not simpler."
> – Albert Einstein



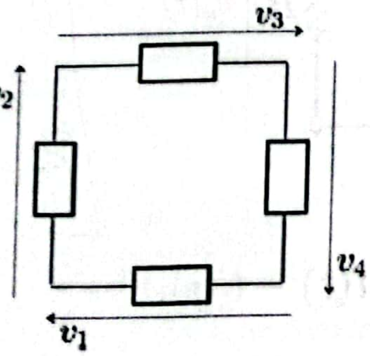
Albert Einstein (1879-1955)

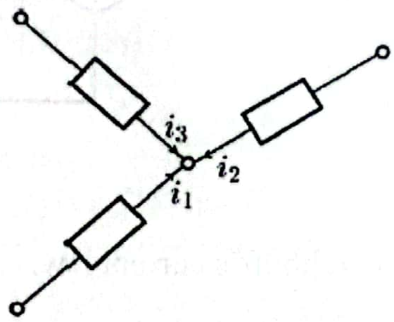
Modeling of Electrical Networks:

- The classical way of writing equations of electrical networks is based on the loop method or the node method,
- Both of which are formulated from the two laws of Kirchhoff, which state:
  - Current law or loop method. The algebraic summation of all currents entering a node is zero. مدخلات التيارات تبديل $\nabla I = 0$
  - Voltage law or node method. The algebraic sum of all voltage drops around a complete closed loop is zero $\Delta V = 0$

Modeling of Electrical Networks:

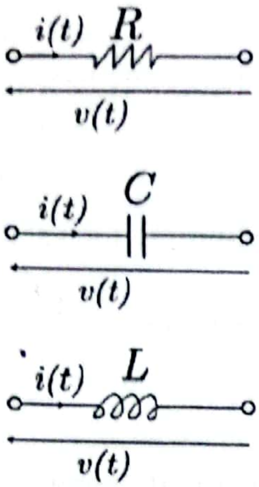
- Kirchhoff's voltage law: balance of voltages on a closed circuit
- Kirchhoff's current law: balance of the currents at a node





$$v_1 + v_2 + v_3 + v_4 = 0$$

$$i_1 + i_2 + i_3 = 0$$



قانون المقاومة

Resistor: $v(t) = Ri(t)$

قانون المكثف

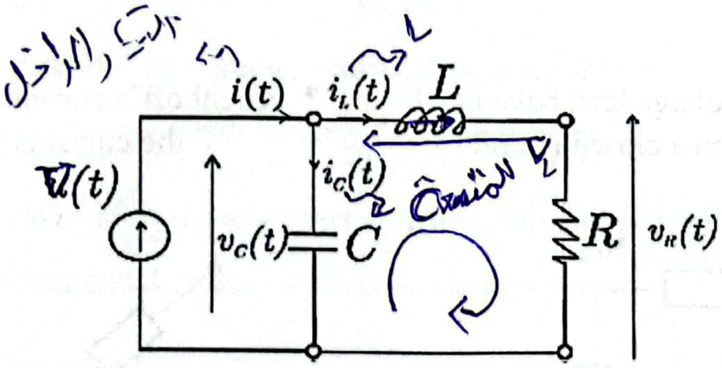
Capacitor: $i(t) = C\frac{dv(t)}{dt}$

قانون الملف

Inductor: $v(t) = L\frac{di(t)}{dt}$

- $v(t)$: voltage across the component
- $i(t)$: current through the component
- R, C, L: resistance, capacitance, inductance

Example of Modeling of Electrical Networks:



① Controle voltage source
② Controle current source.

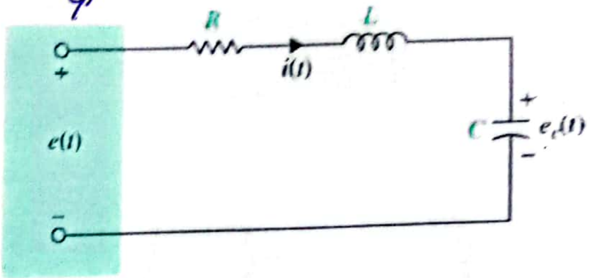
- Kirchhoff's current law: $i_C(t) = C\frac{dv_C(t)}{dt} = i(t) - i_L(t)$
- Kirchhoff's voltage law: $v_c(t) - L\frac{di_L(t)}{dt} - Ri_L(t) = 0$

$\rightarrow V_C(t) - V_L(t) - V_R(t) = 0$

فرق الجهد بين نقطتين $V(t)$ ΔV

Example of Modeling of Electrical Networks:

$$e(t)=e_R+e_L+e_c$$



where e_R = Voltage across the resistor R

e_L = Voltage across the inductor L

e_c = Voltage across the capacitor C

Or

$$e(t)=+e_c(t)+\underbrace{Ri(t)}_{e_R}+\underbrace{L\frac{di(t)}{dt}}_{e_L}$$

Using current in C,

$$C\frac{de_c(t)}{dt}=i(t)$$

$$LC\frac{d^2e_c(t)}{dt^2}+RC\frac{de_c(t)}{dt}+e_c(t)=e(t)$$

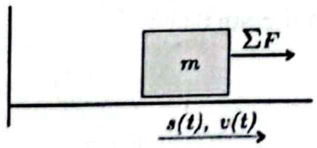
$$LC\frac{d^2e_c(t)}{dt^2}+RC\frac{de_c(t)}{dt}+e_c(t)=e(t)$$

$$\ddot{e}_c(t)+\frac{R}{L}\dot{e}_c(t)+\frac{1}{LC}e_c(t)=\frac{1}{LC}e(t)$$

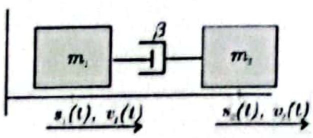
(LC)

Pass

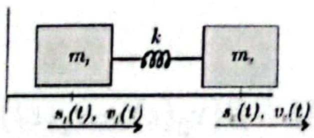
Mechanical systems – Linear motion



Newton's Law:
$$\sum F(t)=m\frac{dv(t)}{dt}=m\frac{d^2s(t)}{dt^2}$$

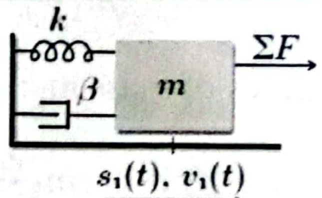


Viscous friction:
$$F_1(t)=\beta(v_2(t)-v_1(t))=-F_2(t)$$



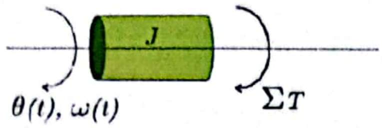
Elastic coupling:
$$F_1(t)=k(s_2(t)-s_1(t))=-F_2(t)$$

- $s_i(t)$, $v_i(t)$ = position and velocity of body i, with respect to a fixed (inertial) reference frame.
- $F_i(t)$ = force acting on body i
- m,β,k = mass, viscous friction coefficient, spring constant

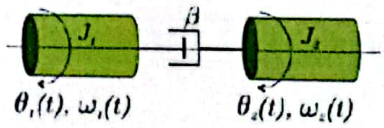


- Special case: $s_2(t)\equiv 0$, $v_2(t)\equiv 0$
  $F_1(t)=-ks_1(t)$, $F_1(t)=-\beta v_1(t)$

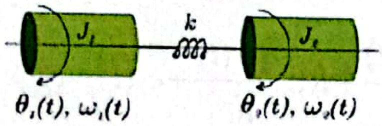
Mechanical systems – Rotational motion



Newton's Law:
$$\sum \tau(t) = J\frac{d\omega(t)}{dt} = J\frac{d^2\theta(t)}{dt^2}$$



Viscous friction:
$$\tau_1(t) = \beta(\omega_2(t) - \omega_1(t)) = -\tau_2(t)$$

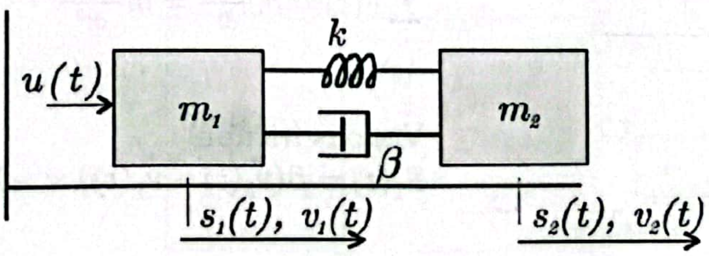


Elastic coupling:
$$\tau_1(t) = k(\theta_2(t) - \theta_1(t)) = -\tau_2(t)$$

- $\theta_i(t)$, $\omega_i(t)$= angular position and angular velocity of body i, with respect to a fixed (inertial) reference frame
- $\tau_i(t)$: torque acting on body i
- J, β, k: inertia, viscous friction coefficient, spring constant

Example of mechanical system

Two masses connected by spring-damper (no dry friction with the surface):



Dynamics of mass m_1:
$$m_1\frac{dv_1(t)}{dt} = u(t) + k(s_2(t) - s_1(t)) + \beta(v_2(t) - v_1(t))$$

Dynamics of mass m_2:
$$m_2\frac{dv_2(t)}{dt} = -k(s_2(t) - s_1(t)) - \beta(v_2(t) - v_1(t))$$

Note: viscous and elastic forces always oppose motion

End of Lectures 2